Adopting A Migration Safety Check Process

Each Active Record Migration file is a new source code file. At organizations that use Pull Requests or Merge Requests to introduce changes, a single change is prepared and a developer reviews it.

Experienced developers might identify dangerous database changes from their past experience, using their familiarity with table row counts and query pattern to notice changes that might cause long lock durations. Raising these concerns can help avoid errors.

On the other end of the spectrum, when Migrations are completely safe to perform at any scale, the authoring should release their change as quickly as possible.

While review feedback from team members is valuable, it requires team members to have skills to identify dangerous migrations and provide safer alternatives. Besides technical challenges, there are social challenges in providing this feedback in a graceful and timely fashion.

A good tool could help identify potentially unsafe Migrations just like a skilled developer and suggest safer alternatives. The tool could be used by any member on the team during their local development process, before publicizing their changes for review.

The tool could automatically block Migrations that are potentially unsafe but also provide an override mechanism.

Fortunately this tool exists and it's called Strong Migrations!

Exploring Strong Migrations Features

Strong Migrations strengthens the uptime and resiliency of Rails apps and engineering teams by adding safety checks for Migrations.

Most unsafe operations are detected and prevented by default. Besides the technical benefit, there is also an educational benefit for developers looking for feedback in a tight feedback loop. Any time Strong Migrations flags a Migration you've created, it's worth carefully reviewing the explanation and suggestions, and even reading more on the GitHub README.

Some of the operations that are detected are listed below.

1. When adding constraints (`CHECK`, `FOREIGN KEY`, and `NOT NULL`) to an existing column, Strong Migrations checks whether they're validated immediately or not